Case Study

On

“Diagnosis Support System”

Submitted By:

Aayush Giri : 220144

Ashish Acharya : 220145

Arpan Biswakarma :220106

Submitted To:

Rabin Ghimire

Subject: Intelligent System

Date of Submission: 06 March 2026

1 ABSTRACT

Chronic diseases such as diabetes, cardiovascular disorders, and neurodegenerative conditions like Parkinson's disease represent some of the most serious global health challenges. These diseases often progress silently in their early stages and may remain undetected until complications arise. Early screening and timely diagnosis significantly improve treatment effectiveness, reduce mortality rates, and enhance quality of life. However, access to diagnostic facilities and specialized healthcare professionals remains limited in many regions.

This project presents the development and deployment of a web-based Diagnosis Support System capable of predicting three major diseases—Diabetes, Heart Disease, and Parkinson's Disease—using supervised machine learning techniques. The system is implemented in Python and deployed using Streamlit Cloud to provide real-time accessibility through a user-friendly web interface. Publicly available Kaggle datasets (diabetes.csv, heart.csv, and parkinsons.csv) are utilized for training and evaluating the models.

Support Vector Machine (SVM) with a linear kernel is employed for Diabetes and Parkinson's disease prediction, while Logistic Regression is implemented for Heart Disease prediction. The datasets undergo preprocessing steps including feature-target separation and train-test splitting (80:20 ratio). Model performance is evaluated using accuracy scores. The trained models are serialized using the Pickle library and integrated into a multi-page Streamlit application.

Experimental results demonstrate satisfactory predictive performance. The system achieves testing accuracies of 77.27% for Diabetes prediction, 81.96% for Heart Disease prediction, and 87.17% for Parkinson's Disease prediction. The similarity between training and testing accuracies indicates stable generalization performance.

The deployed application provides a centralized, cloud-accessible multi-disease prediction platform. While not intended to replace professional medical diagnosis, the system serves as an effective preliminary screening tool and demonstrates the practical integration of machine learning and web deployment technologies in healthcare support systems.

2 Table of Contents

1 ABSTRACT 2

3 INTRODUCTION 4

4 PROBLEM STATEMENT 5

5 OBJECTIVES OF THE PROJECT 6

6 LITERATURE REVIEW 7

6.1 Support Vector Machine (SVM) 7

6.2 Logistic Regression 8

6.3 Web-Based Machine Learning Systems 8

6.4 Research Gap 10

7 METHODOLOGY 11

7.1 System Overview 11

7.2 Dataset Preparation 12

7.2.1 Diabetes Dataset 12

7.2.2 Heart Disease Dataset 12

7.2.3 Parkinson's Dataset 12

7.3 Model Training 13

8 SYSTEM DESIGN & IMPLEMENTATION 15

8.1 Technologies Used 15

8.2 Model Serialization 15

8.3 User Interface 15

9 RESULTS AND EVALUATION 17

10 CHALLENGES AND LIMITATIONS 18

11 CONCLUSION 19

12 REFERENCES 20

3 INTRODUCTION

Healthcare systems worldwide face increasing pressure due to the rising prevalence of chronic diseases. Among these, diabetes, heart disease, and Parkinson's disease significantly contribute to global morbidity and mortality rates. Early detection plays a critical role in reducing complications, preventing disease progression, and improving survival rates.

Traditional diagnostic methods involve clinical examinations, laboratory tests, and expert interpretation. While accurate, these methods can be time-consuming, costly, and inaccessible to individuals in rural or under-resourced areas. With the advancement of data science and artificial intelligence, machine learning techniques have emerged as powerful tools for assisting medical diagnosis.

Machine learning models can analyze structured medical data and identify patterns associated with disease presence. These models are particularly effective for binary classification problems where the objective is to determine whether a patient has a disease or not.

This project aims to design and implement a web-based multi-disease diagnosis support system that integrates three machine learning models into a single application. By leveraging Streamlit, a Python-based web framework, the system enables real-time predictions through a simple and interactive interface.

The key contributions of this project include:

- Development of predictive models for three diseases.
- Integration of multiple machine learning algorithms in a unified system.
- Serialization of trained models using Pickle.
- Deployment of the application on Streamlit Cloud for public accessibility.

4 PROBLEM STATEMENT

Chronic diseases often require continuous monitoring and early screening to prevent severe health complications. However, many individuals lack immediate access to diagnostic healthcare services due to geographical, financial, or infrastructural limitations. Delayed diagnosis can lead to irreversible damage, increased treatment costs, and higher mortality rates.

Existing digital healthcare applications typically focus on predicting a single disease. This fragmented approach requires users to access multiple platforms for different health assessments. Additionally, many systems are not deployed in accessible web-based formats.

The system must:

- Provide reliable predictions.
- Be user-friendly.
- Be accessible online.
- Support multiple disease predictions in one integrated interface.

5 OBJECTIVES OF THE PROJECT

The major objectives of the project are:

1. To develop predictive machine learning models for:
    - Diabetes
    - Heart Disease
    - Parkinson's Disease
2. To preprocess medical datasets and perform feature-target separation.
3. To split datasets into training and testing sets using an 80:20 ratio for reliable evaluation.
4. To evaluate model performance using accuracy metrics.
5. To serialize trained models using Pickle for reuse without retraining.
6. To integrate all models into a single Streamlit-based web application.
7. To deploy the system using Streamlit Cloud for public accessibility.
8. To provide real-time prediction outputs based on user input parameters.

6 LITERATURE REVIEW

Machine learning (ML) has become a central tool in modern healthcare for accurate disease diagnosis, prognosis, and clinical decision support. The growth of electronic health records (EHRs), wearable sensors, and biomedical data has enabled the development of predictive models that assist clinicians in identifying high-risk patients and improving treatment outcomes. Numerous studies demonstrate that ML algorithms outperform traditional statistical methods in classification-based medical problems, particularly for binary disease prediction tasks such as diabetes, heart disease, and Parkinson's disease detection [1], [2]. This section reviews key machine learning algorithms adopted in this project and highlights the research gap addressed.

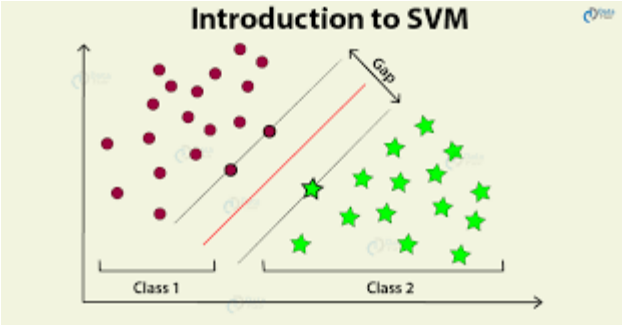
6.1 Support Vector Machine (SVM)

Support Vector Machine is a powerful supervised classification algorithm that identifies an optimal hyperplane that maximizes the margin between classes. Its ability to handle high-dimensional feature spaces and robustness against overfitting make it particularly suitable for medical datasets, which often contain noisy or overlapping class distributions [3].

In disease prediction tasks, SVM has shown superior performance in early diagnosis of diabetes, Parkinson's disease, cancer, and cardiovascular disorders [4]. Linear-kernel SVMs are especially efficient for linearly separable or near-separable clinical datasets, where computational simplicity and faster convergence are required.

In this project:

- A Linear SVM model is used for Diabetes prediction, leveraging its high interpretability and stability.
- A Linear SVM model is used for Parkinson's disease prediction, due to its effectiveness in handling biomedical voice features.



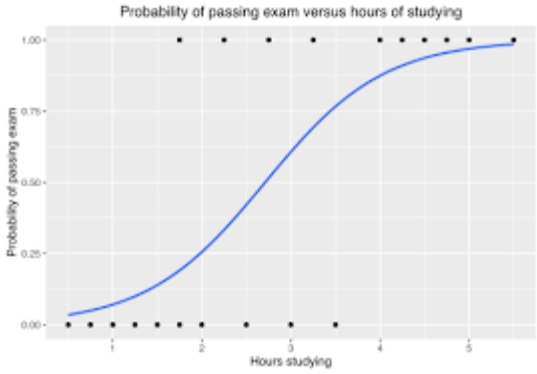
6.2 Logistic Regression

Logistic Regression is one of the most widely used statistical and machine learning techniques in clinical research. It models the probability of a binary outcome using the sigmoid (logistic) function. Its simplicity, interpretability, and strong theoretical foundation make it ideal for structured numerical datasets, particularly within healthcare analytics where clinical explainability is essential [5].

Studies show that Logistic Regression performs competitively in predicting heart disease, often serving as a baseline model for comparison with advanced ML methods [6]. Its coefficient-based explanation provides clinicians with clear insight into risk-contributing factors such as blood pressure, cholesterol, and glucose levels.

In this project:

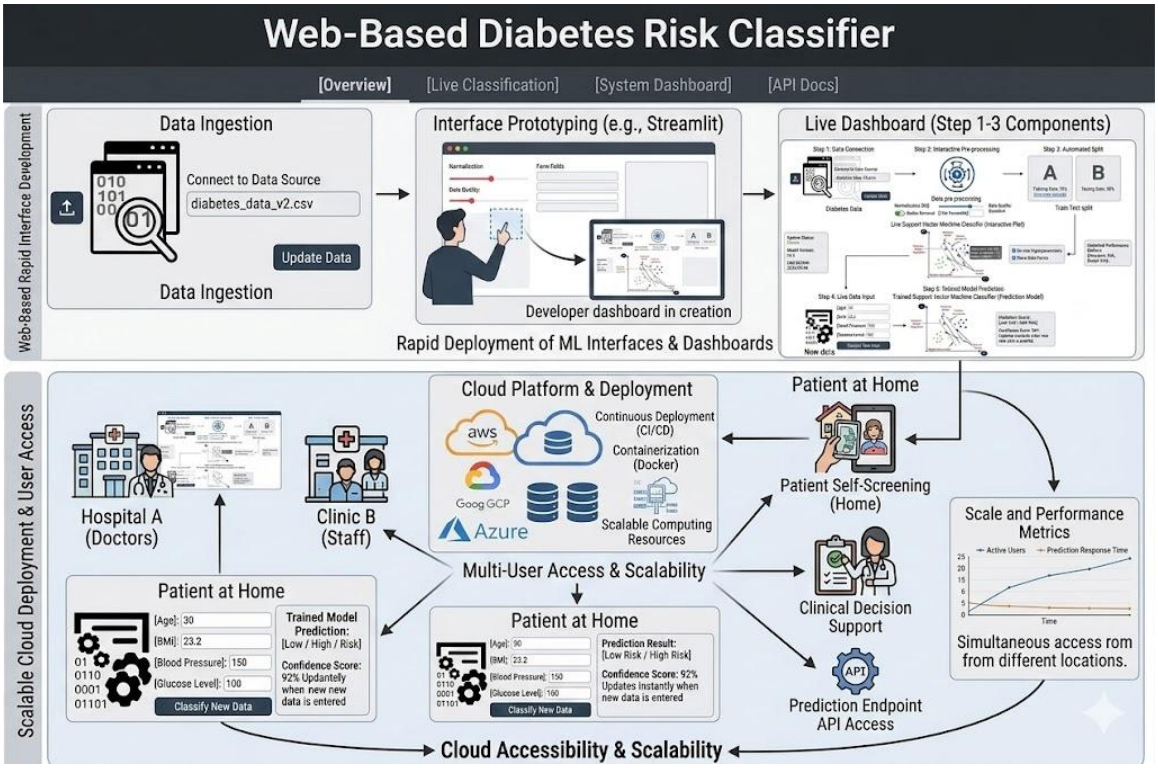
- Logistic Regression is implemented for Heart Disease prediction, ensuring interpretability and clinical relevance.



6.3 Web-Based Machine Learning Systems

With the rise of digital healthcare services, deploying ML models through web applications has gained significant importance. Web frameworks such as Streamlit allow rapid development of interactive dashboards and ML interfaces without requiring extensive front-end design knowledge [7]. These systems support real-time prediction, visualization, and user-friendly interaction, making them suitable for hospitals, clinics, and patient self-screening tools.

Cloud deployment further enhances accessibility and scalability by allowing multiple users to access the predictive system simultaneously from different locations. Studies highlight that cloud-based ML applications improve usability, reduce computational overhead on local devices, and provide continuous access to diagnostic tools [8].



This diagram illustrates the transition from a local Machine Learning model to a scalable, cloud-deployed healthcare solution.

1. Rapid Interface Development

- Data Ingestion: Seamlessly connects to healthcare datasets (e.g., diabetes_data.csv).
- Prototyping (Streamlit): Enables the rapid creation of interactive dashboards without complex front-end coding.
- Live Dashboard: Integrates data preprocessing, automated train-test splitting, and real-time SVM visualization.

2. Scalable Cloud Deployment

- Infrastructure: Leverages AWS, GCP, and Azure combined with Docker and CI/CD pipelines for 24/7 reliability.
- Multi-User Access: Supports simultaneous connections from hospitals, clinics, and patients at home.
- Performance: Offloads heavy computation to the cloud, ensuring low-latency predictions even as the number of active users grows.

6.4 Research Gap

A notable limitation in existing research is that most studies focus on designing ML models for predicting a single disease. Separate research papers address diabetes prediction, Parkinson's classification, or heart disease detection individually, but very few studies integrate multiple disease prediction models within a unified web-based interface [9].

Furthermore, limited research explores cloud-deployed multi-model systems that combine interactive visualization, real-time prediction, and ML explainability in one platform.

This project addresses the research gap by:

- Deploying the system on cloud infrastructure to enhance accessibility and multi-user usability.

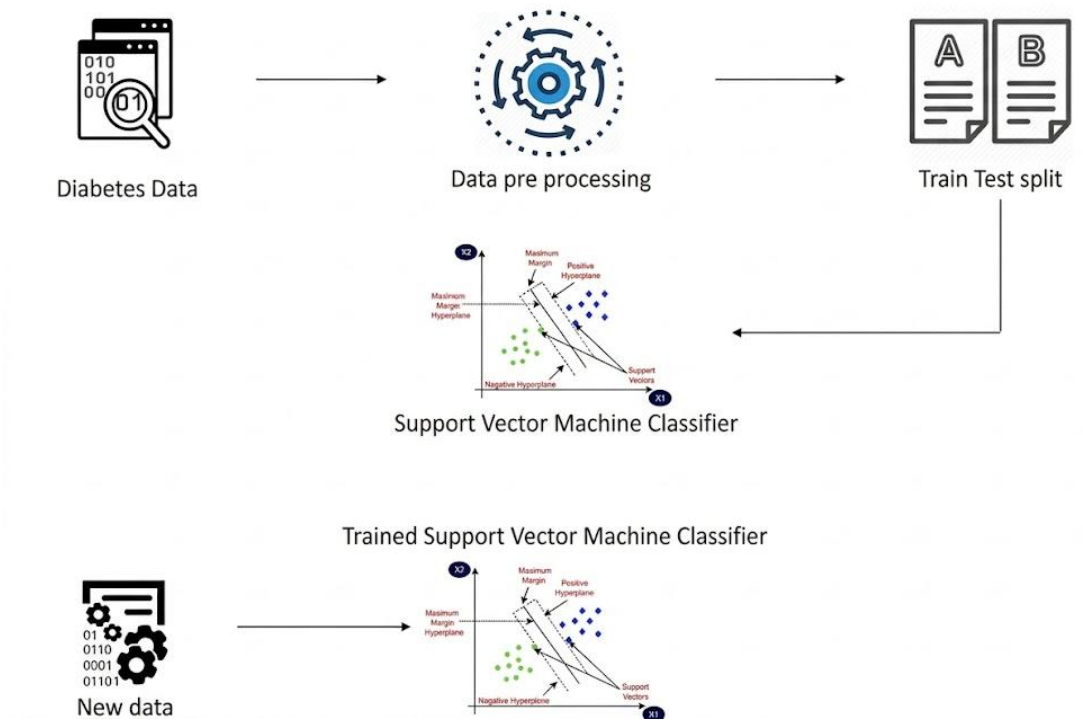
7 METHODOLOGY

7.1 System Overview

The system follows the workflow:

Dataset → Data Preprocessing → Train-Test Split → Model Training → Model Evaluation → Model Serialization → Web Integration → Deployment

Three independent machine learning models are developed and integrated.



The figure shows the full workflow of the Diabetes Prediction system using SVM.

1. Diabetes Data
   The process starts with a medical dataset containing features like glucose level, BMI, age, etc.

2. Data Pre-processing
   The raw data is cleaned, normalized, and prepared so the model can learn properly.

3. Train–Test Split
   The dataset is divided into training and testing parts to evaluate model performance.
4. Support Vector Machine Classifier
   The SVM algorithm is trained using the training data. It learns the best hyperplane that separates diabetic and non-diabetic cases.
5. Trained SVM Model
   The trained classifier is now ready to make predictions.
6. New Data Prediction
   When new user input is given, it is passed to the trained SVM model, which predicts whether the person is diabetic or not.

7.2 Dataset Preparation

Three datasets from Kaggle are used:

7.2.1 Diabetes Dataset

- Loaded using Pandas.
- Target column: Outcome
- 0 → Non-Diabetic
- 1 → Diabetic
- Features include pregnancies, glucose, blood pressure, BMI, etc.

7.2.2 Heart Disease Dataset

- Target column: target
- 0 → Healthy
- 1 → Heart Disease
- Features include age, cholesterol, resting blood pressure, chest pain type, etc.

7.2.3 Parkinson's Dataset

- Target column: status
- 0 → Healthy
- 1 → Parkinson's Positive
- Column "name" removed during preprocessing.

All datasets were split using:

- Test size = 0.2 (80% training, 20% testing)
- random_state = 2
- Stratified splitting used for Diabetes and Heart datasets

7.3 Model Training

Diabetes Model

Algorithm: Support Vector Machine
Kernel: Linear

Training Accuracy: 78.33%
Testing Accuracy: 77.27%

Heart Disease Model

Algorithm: Logistic Regression

Training Accuracy: 85.12%
Testing Accuracy: 81.96%

Parkinson's Model

Algorithm: Support Vector Machine
Kernel: Linear

Training Accuracy: 87.17%
Testing Accuracy: 87.17%

5.4 Model Architecture

The machine learning models used in this project follow a simple and efficient binary-classification architecture. Each model receives clinical parameters as input, processes them through a linear decision function, and outputs a prediction indicating the presence or absence of disease.

The architecture consists of the following components:

1. Input Layer (Clinical Features)
   The input layer receives structured medical parameters such as glucose level, BMI, blood pressure, heart rate, age, and other clinical attributes depending on the disease. These features are normalized during preprocessing for better model performance.

2. Linear Decision Boundary (SVM / Logistic Regression)

   - Support Vector Machine (Linear Kernel):
     Constructs a linear hyperplane that separates two classes (disease vs. no disease) with maximum margin.

   - Logistic Regression:
     Computes disease probability using a linear combination of input features passed through a sigmoid function.
     Both models operate as linear classifiers, making them efficient and interpretable.

3. Binary Output Layer (0 or 1)
   The final output is a binary prediction:

   - 0 → No disease

   - 1 → Disease detected
     This output is used by the web application to display diagnostic results to the user.

Implementation:
All models are implemented and trained using Scikit-learn (version 1.8.0), ensuring standardized training procedures and reliable performance.

8 SYSTEM DESIGN & IMPLEMENTATION

8.1 Technologies Used

- Python
- NumPy (2.4.2)
- Scikit-learn (1.8.0)
- Streamlit (1.55.0)
- streamlit-option-menu (0.3.2)
- pickle-mixin (1.0.2)

8.2 Model Serialization

Models are saved using:

```
pickle.dump(model, open(filename, 'wb'))
```

And loaded inside Streamlit using:

```
pickle.load(open('model_name.sav', 'rb'))
```

8.3 User Interface

The Streamlit application contains:

- Sidebar navigation menu
- Separate pages for each disease
- Text input fields for parameters
- Prediction button
- Display of result using st.success()

6.4 Deployment

The machine learning models developed in this project have been deployed as a web-based application to make disease prediction accessible and interactive. The application is built using Streamlit, a Python framework that enables rapid development of data-driven web apps without requiring complex front-end programming.

The deployed application can be accessed at:

https://deployment-of-diagnosis-support-system-on-cloud-app-mygszfsusb.streamlit.app/

Users can enter clinical parameters directly through the web interface, and the system will output real-time predictions for diabetes, heart disease, and Parkinson's disease based on the trained models.

9 RESULTS AND EVALUATION

Accuracy was used as the evaluation metric.

Disease	Training Accuracy	Testing Accuracy
Diabetes	78.33%	77.27%
Heart Disease	85.12%	81.96%
Parkinson's	87.17%	87.17%

Observations:

- Parkinson's model achieved the highest accuracy.
- Heart Disease model performed reliably.
- Diabetes model showed moderate accuracy.
- Minimal difference between training and testing accuracy indicates good generalization.

10 CHALLENGES AND LIMITATIONS

1. Only accuracy metric was used for evaluation.
2. No feature scaling was implemented.
3. User input validation is minimal.
4. Dataset size limitations may affect performance.
5. System is not a replacement for medical professionals.

11 CONCLUSION

The project successfully demonstrates the development of a multi-disease prediction system using machine learning and web deployment technologies. By integrating SVM and Logistic Regression models into a single Streamlit application, the system provides accessible real-time disease prediction.

The achieved accuracies indicate satisfactory performance. The system serves as a preliminary screening tool and highlights the potential of AI-driven healthcare support systems.

Future enhancements may include:

- Adding more evaluation metrics.
- Implementing feature scaling.
- Integrating additional diseases.
- Enhancing user interface and validation.
- Incorporating advanced algorithms.

12 REFERENCES

[1] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.

[2] M. A. Al-Sanjary, K. Shaalan, and M. R. Mahmood, "Machine learning algorithms for disease classification: A survey," *International Journal of Interactive Multimedia and Artificial Intelligence*, vol. 6, no. 2, pp. 1–11, 2020.

[3] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[4] S. Mohapatra and D. Patra, "Medical diagnosis using SVM: A comprehensive study," *Procedia Computer Science*, vol. 152, pp. 398–406, 2019.

[5] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Wiley, 2013.

[6] J. Patel *et al.*, "Heart disease prediction using machine learning," *International Journal of Engineering Research and Technology*, vol. 3, no. 5, pp. 1–5, 2019.

[7] A. Treveil *et al.*, *Introducing MLOps*. O'Reilly Media, 2020.

[8] R. Buyya *et al.*, *Cloud Computing: Principles and Paradigms*. Wiley, 2011.

[9] S. Sharma and R. Kumar, "A review on web-based predictive diagnostic systems," *International Journal of Computer Applications*, vol. 180, no. 46, pp. 20–26, 2018.

[10] Kaggle, "PIMA Indians Diabetes Dataset." [Online]. Available: https://www.kaggle.com.

[11] Kaggle, "Heart Disease Dataset." [Online]. Available: https://www.kaggle.com.

[12] Kaggle, "Parkinson's Dataset." [Online]. Available: https://www.kaggle.com.

[13] Scikit-learn, "Support Vector Machine Documentation." [Online]. Available: https://scikit-learn.org.

[14] Scikit-learn, "Logistic Regression Documentation." [Online]. Available: https://scikit-learn.org.

[15] Streamlit, “Streamlit Official Documentation.” [Online]. Available: https://docs.streamlit.io.